Lecture 25

The Riemann integral and uniform continuity

Uses.

Summary 1: The number $\sqrt{2}$ is irrational.
Summary 2: Induction is available to prove the finite-square-sum identity below. Density supplies rationals in every real interval. The Archimedean property supplies fine meshes. R supplies infima and suprema of nonempty bounded real sets. Absolute values satisfy the triangle and reverse triangle inequalities.
Summary 3: The real limit laws control finite algebraic expressions.
Summary 4: Eventual order passes to limits. The squeeze theorem proves convergence. Every bounded real sequence has a convergent subsequence; closed-interval bounds survive its limit.
Summary 5: Continuity is relative to the domain. A continuous function preserves the limit of every approaching domain sequence.

Objectives. *We define an integral by lower and upper estimates on a finite interval. Their difference gives a criterion for integrability. We prove that monotone functions satisfy it, then use subsequences to obtain uniform continuity for continuous functions on a closed interval.*

By the end of this lecture you will be able to:

- Construct Darboux sums and use their gap to test integrability.
- Prove integrability of a monotone function on a closed finite interval.
- Use integral linearity, order and the absolute-value bound.
- Prove uniform continuity for a continuous function on a closed finite interval.

25.1 Lower and upper sums

Fix real numbers $a < b$. A *partition* of $[a, b]$ is a finite list

$$P: \quad a = x_0 < x_1 < \cdots < x_n = b.$$

Its subinterval lengths are $\Delta x_i = x_i - x_{i-1} > 0$. Their sum is $b - a$. Its *mesh* is the largest length, $\|P\| = \max_i \Delta x_i$.

Let $f : [a, b] \to \mathbb{R}$ be bounded. On each subinterval define

$$m_i = \inf\{f(x) : x \in [x_{i-1}, x_i]\}, \qquad M_i = \sup\{f(x) : x \in [x_{i-1}, x_i]\}.$$

These numbers exist by R, applied also to negatives for infima. They need not be values attained by f. The definitions therefore use infima and suprema, rather than assume the existence of minima and maxima.

Definition 25.1 (Darboux sums)**.** The *lower sum* and *upper sum* of a bounded real function f for a partition P are

$$L(P,f)=\sum_{i=1}^{n} m_i\Delta x_i, \qquad U(P,f)=\sum_{i=1}^{n} M_i\Delta x_i.$$

The symbol $\sum_{i=1}^{n}$ denotes addition of the terms indexed from 1 to n.

For a constant function $f(x)=c$, both sums equal $c(b-a)$ for every partition. For a nonnegative function, the lower rectangles lie below the graph and the upper rectangles lie above it. The sums also work when f has negative values. They are then signed quantities, not unsigned areas.

For a finite calculation, take $f(x)=x$ on $[0,1]$ and $P=(0,1/4,1)$. The two cell widths are $1/4$ and $3/4$. The infima are $0, 1/4$ and the suprema are $1/4, 1$, so

$$L(P,f)=0\cdot\tfrac{1}{4}+\tfrac{1}{4}\cdot\tfrac{3}{4}=\tfrac{3}{16}, \qquad U(P,f)=\tfrac{1}{4}\cdot\tfrac{1}{4}+1\cdot\tfrac{3}{4}=\tfrac{13}{16}.$$

Choose one sample point in each cell, $1/8$ and $5/8$; these are called tags. Their width-weighted sum is $(1/8)(1/4)+(5/8)(3/4)=1/2$, between these bounds. This finite comparison does not by itself establish the integral; it is the starting point for refining partitions and controlling the gap.

A *refinement* Q of P retains every point of P and may add points. On each smaller subinterval its infimum is at least the old infimum and its supremum is at most the old supremum. The smaller lengths sum to the old length. Adding over the old subintervals therefore gives

$$L(P,f)\le L(Q,f)\le U(Q,f)\le U(P,f). \tag{25.1}$$

Two unrelated partitions also have comparable bounds. Their union of points is a common refinement R, so

$$L(P,f)\le L(R,f)\le U(R,f)\le U(Q,f).$$

Thus every lower sum is at most every upper sum, even when the partitions differ. Refinement, rather than the number of subintervals alone, justifies this comparison.

Definition 25.2 (Riemann integral)**.** For bounded $f:[a,b]\to\mathbb{R}$, define the lower and upper integrals by

$$\underline{I}(f)=\sup_P L(P,f), \qquad \overline{I}(f)=\inf_P U(P,f).$$

The function is *Riemann integrable* if these numbers agree. Their common value is denoted by

$$\int_a^b f(x)\,\mathrm{d}x.$$

If $|f(x)|\le M$, all sums lie between $-M(b-a)$ and $M(b-a)$. Both collections of sums are nonempty. Their infimum and supremum are therefore finite, and the preceding comparison gives $\underline{I}(f)\le\overline{I}(f)$. For the constant function, the definition gives $\int_a^b c\,\mathrm{d}x=c(b-a)$.

25.2 The Darboux criterion and a computed integral

Theorem 25.3 (Darboux criterion). *A bounded function $f : [a, b] \to \mathbb{R}$ is Riemann integrable if and only if for every $\varepsilon > 0$ some partition P satisfies*

$$U(P, f) - L(P, f) < \varepsilon.$$

Proof. First suppose the two integrals equal I. By the definition of supremum, choose P_1 with $L(P_1, f) > I - \varepsilon/2$. By the definition of infimum, choose P_2 with $U(P_2, f) < I + \varepsilon/2$. Their common refinement P satisfies both estimates by Eq. (25.1). Subtracting gives the required gap.

Conversely, for each such partition,

$$0 \le \overline{I}(f) - \underline{I}(f) \le U(P, f) - L(P, f) < \varepsilon.$$

A nonnegative real number smaller than every positive ε is zero. Hence the two integrals agree. ∎

Choose a *tag* $t_i \in [x_{i-1}, x_i]$ in each subinterval. The corresponding *Riemann sum* is

$$S(P, f, t) = \sum_{i=1}^{n} f(t_i)\Delta x_i.$$

Its terms lie between the corresponding lower and upper terms, so $L(P, f) \le S(P, f, t) \le U(P, f)$. Once f is integrable with value I, both S and I lie in that same interval. Thus $|S - I| \le U(P, f) - L(P, f)$. Along partitions whose Darboux gaps tend to zero, every choice of tags therefore gives sums tending to I. The needed estimate is the gap; we have not assumed it merely from a small mesh.

Figure 25.1 shows the estimates for $f(x) = x^2$.

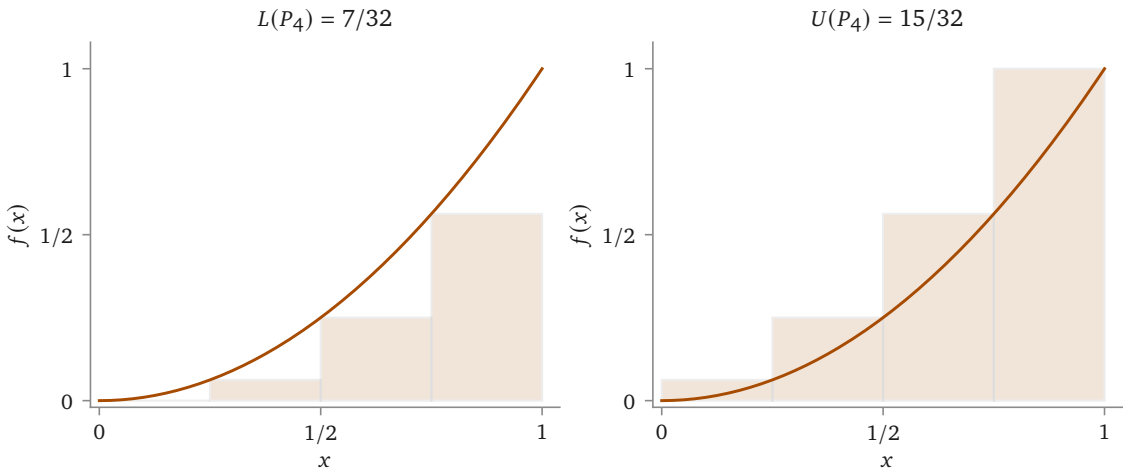


Figure 25.1: Lower and upper rectangles for x^2 on $[0, 1]$ with four equal subintervals. The same partition gives the bounds $7/32$ and $15/32$.

A real function $f : A \to \mathbb{R}$, with $A \subseteq \mathbb{R}$, is *nondecreasing* if $x \le y$ implies $f(x) \le f(y)$ for every $x, y \in A$. It is *nonincreasing* if the output inequality is reversed. It is *monotone* if it has either property.

Example 25.4 (The integral of a square)**.** Let $f(x) = x^2$ on $[0, 1]$. Use the uniform partition P_n with $x_i = i/n$. Squaring is nondecreasing on this interval, so the extreme values in the ith subinterval occur at its endpoints. Hence

$$L(P_n, f) = \frac{1}{n^3} \sum_{k=0}^{n-1} k^2, \qquad U(P_n, f) = \frac{1}{n^3} \sum_{k=1}^{n} k^2.$$

The finite square sum is $F(n) = n(n+1)(2n+1)/6$. Indeed, $F(1) = 1$ and expansion gives $F(n+1) - F(n) = (n+1)^2$. Induction therefore proves the formula; the empty sum at zero is zero. Substitution gives

$$L(P_n, f) = \frac{(n-1)(2n-1)}{6n^2}, \qquad U(P_n, f) = \frac{(n+1)(2n+1)}{6n^2}.$$

Their difference is $1/n$, so the criterion proves integrability. Both sums tend to $1/3$ by the limit laws. Since the integral lies between them, order and the squeeze theorem give

$$\int_0^1 x^2 \, \mathrm{d}x = \frac{1}{3}.$$

25.3 Monotone functions and a failure of integrability

Theorem 25.5 (Monotone integrability)**.** *Every monotone real function on $[a, b]$ is Riemann integrable.*

Proof. Suppose first that f is nondecreasing. Its endpoint values bound it. For the uniform partition into n intervals, the endpoints give the infimum and supremum on each subinterval. The difference of the sums is

$$U(P_n, f) - L(P_n, f) = \frac{b-a}{n} \sum_{i=1}^{n} \bigl(f(x_i) - f(x_{i-1})\bigr) = \frac{b-a}{n} \bigl(f(b) - f(a)\bigr).$$

The sum cancels consecutive intermediate values. The gap tends to zero, so the Darboux criterion applies. For nonincreasing f, the endpoint roles reverse and the same calculation gives $(b-a)(f(a) - f(b))/n \to 0$. ∎

Boundedness alone is insufficient. Consider the function

$$D(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \notin \mathbb{Q}, \end{cases} \qquad x \in [a, b].$$

Every subinterval of positive length contains a rational number, by Lecture 2. It also contains an irrational number: choose a rational r between its endpoints shifted by $-\sqrt{2}$. Then $r + \sqrt{2}$ lies in the original interval and is irrational, since otherwise subtracting r would make $\sqrt{2}$ rational.

Consequently every subinterval has infimum zero and supremum one. For every partition, $L(P, D) = 0$ and $U(P, D) = b - a$. The gap cannot be made arbitrarily small, so D is not integrable. This is a bounded function for which refinement never reduces the gap.

25.4 Properties of the integral

Theorem 25.6 (Linearity, order and absolute values)**.** *Let f and g be Riemann integrable on $[a,b]$, and let $c \in \mathbb{R}$. The functions $f+g$, cf and $|f|$ are integrable, with*

$$\int_a^b (f+g)\,\mathrm{d}x = \int_a^b f\,\mathrm{d}x + \int_a^b g\,\mathrm{d}x, \qquad \int_a^b cf\,\mathrm{d}x = c\int_a^b f\,\mathrm{d}x.$$

If $f(x) \le g(x)$ for all $x \in [a,b]$, then $\int_a^b f \le \int_a^b g$. Moreover,

$$\left|\int_a^b f\,\mathrm{d}x\right| \le \int_a^b |f|\,\mathrm{d}x.$$

Proof of linearity. On each subinterval, the sum of the two infima is a lower bound for $f+g$, and the sum of the two suprema is an upper bound. Hence

$$L(P,f)+L(P,g) \le L(P,f+g) \le U(P,f+g) \le U(P,f)+U(P,g).$$

Given $\varepsilon > 0$, choose partitions for f and g with gaps below $\varepsilon/2$, and take their common refinement. The displayed estimates give a gap below ε for $f+g$, proving its integrability. They also put both $\int(f+g)$ and $\int f + \int g$ between $L(P,f)+L(P,g)$ and $U(P,f)+U(P,g)$. The interval's width is below ε. Since ε was arbitrary, the two values agree.

For $c > 0$, lower and upper sums are multiplied by c. For $c < 0$, multiplication reverses the extrema: $L(P,cf) = cU(P,f)$ and $U(P,cf) = cL(P,f)$. In either case the gap is multiplied by $|c|$, and the same bounding argument gives $\int cf = c\int f$. For $c = 0$, this is the constant-zero integral. ∎

Proof of order and the absolute-value bound. If $f \le g$, then $L(P,f) \le L(P,g) \le \int_a^b g$ for every P. Taking the supremum over P proves order.

To prove integrability of $|f|$, fix a subinterval with bounds m_i and M_i for f. For any two points x, y in it, the reverse triangle inequality gives

$$\big||f(x)| - |f(y)|\big| \le |f(x)-f(y)| \le M_i - m_i.$$

Taking a supremum over x and then an infimum over y shows that the oscillation (supremum minus infimum) of $|f|$ is at most $M_i - m_i$. Thus its Darboux gap is no larger than that of f, and the criterion proves integrability. Finally, $-|f| \le f \le |f|$. Order and the negative-scalar formula give the stated absolute-value bound. ∎

The integral is also additive over adjacent intervals. For $a < c < b$, a bounded $f : [a,b] \to \mathbb{R}$ is integrable exactly when both restrictions to $[a,c]$ and $[c,b]$ are integrable. In that case,

$$\int_a^b f\,\mathrm{d}x = \int_a^c f\,\mathrm{d}x + \int_c^b f\,\mathrm{d}x \tag{25.2}$$

Indeed, adjoining c to a small-gap partition gives small gaps for both restricted intervals, since their nonnegative gaps add to the full gap. Conversely, joining partitions of the two intervals adds their lower sums and their upper sums. Taking partitions with gaps tending to zero proves the displayed identity. This also proves the reverse integrability assertion if both restricted functions are integrable.

25.5 Continuity and a uniform estimate

Recall from Definition 5.8 that for $f : A \to \mathbb{R}$ on $A \subseteq \mathbb{R}$, continuity at $c \in A$ means that for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$x \in A, \quad |x - c| < \delta \quad \Longrightarrow \quad |f(x) - f(c)| < \varepsilon.$$

The function is *continuous on* A if this holds at each $c \in A$. By Proposition 5.9, if $x_n \in A$ and $x_n \to c \in A$, continuity gives $f(x_n) \to f(c)$: choose δ for a requested error, then take n large enough that $|x_n - c| < \delta$. This uses only the sequence-limit definition.

Definition 25.7 (Uniform continuity)**.** A function $f : A \to \mathbb{R}$ is *uniformly continuous* if

$$\forall \varepsilon > 0 \quad \exists \delta > 0 \quad \forall x, y \in A : \quad |x - y| < \delta \implies |f(x) - f(y)| < \varepsilon.$$

Here a single δ works for all pairs in A. In pointwise continuity, the allowed δ may depend on the point c. A uniform choice also works at each fixed c, so uniform continuity implies continuity.

Theorem 25.8 (Heine–Cantor)**.** *Every continuous real function on $[a, b]$ is uniformly continuous.*

Proof. Suppose uniform continuity failed. Negating its definition gives a fixed $\varepsilon_0 > 0$ for which every $\delta > 0$ has a failing pair. With $\delta = 1/n$, choose $x_n, y_n \in [a, b]$ such that

$$|x_n - y_n| < \frac{1}{n}, \qquad |f(x_n) - f(y_n)| \geq \varepsilon_0.$$

Choose a subsequence $x_{n_k} \to c \in [a, b]$ by Bolzano–Weierstrass. The triangle estimate

$$|y_{n_k} - c| \leq |y_{n_k} - x_{n_k}| + |x_{n_k} - c| < \frac{1}{n_k} + |x_{n_k} - c|$$

shows that $y_{n_k} \to c$ too, by the squeeze theorem and $n_k \geq k$. Continuity at this same point gives both image subsequences limit $f(c)$. For large k, each error from $f(c)$ is below $\varepsilon_0/3$. The triangle inequality then gives

$$|f(x_{n_k}) - f(y_{n_k})| < \frac{2\varepsilon_0}{3},$$

contradicting the chosen failing-pair inequality. ∎

The closed-interval hypothesis matters. The function $f(x) = 1/x$ is continuous on $(0, 1]$: near $c > 0$, require $|x - c| < c/2$, so $x > c/2$ and $|1/x - 1/c| \leq 2|x - c|/c^2$. The choice $\delta = \min\{c/2, \varepsilon c^2/2\}$ supplies the requested error. It is not uniformly continuous there. For any proposed $\delta > 0$, put $t = \min\{\delta/2, 1/2\}$. The points t and $t/2$ belong to $(0, 1]$ and have distance $t/2 < \delta$, while their function values differ by $1/t \geq 2$. Thus the fixed tolerance 1 fails for every proposed uniform choice.

Summary 25.

- Definitions 25.1 and 25.2 and Theorem 25.3: a bounded real function on $[a, b]$, $a < b$, is integrable exactly when its Darboux gap can be made arbitrarily small.
- Theorem 25.5: every monotone real function on $[a, b]$ is integrable. The bounded Dirichlet function is not integrable.
- Theorem 25.6 and Eq. (25.2): integrals are linear, order-preserving and additive over adjacent intervals; $|\int f| \le \int |f|$.
- Definition 25.7 and Theorem 25.8: a continuous real function on a closed finite interval is uniformly continuous. The proof uses Bolzano–Weierstrass from Lecture 4.

Lecture 26 proves integrability of continuous functions before connecting this integral with antiderivatives.

References

R. G. Bartle and D. R. Sherbert, *Introduction to Real Analysis*, 4th ed., Wiley, 2011, §§5.4 and 7.1–7.2. S. R. Ghorpade and B. V. Limaye, *A Course in Calculus and Real Analysis*, 2nd ed., Springer, 2018, §§6.1–6.2.